

Introduction to NLP :

Natural Language Processing (NLP) is a subfield of artificial intelligence that gives computers the ability to read, understand, and generate human language .

It combines computer science and linguistics to bridge the gap between human communication and machine understanding

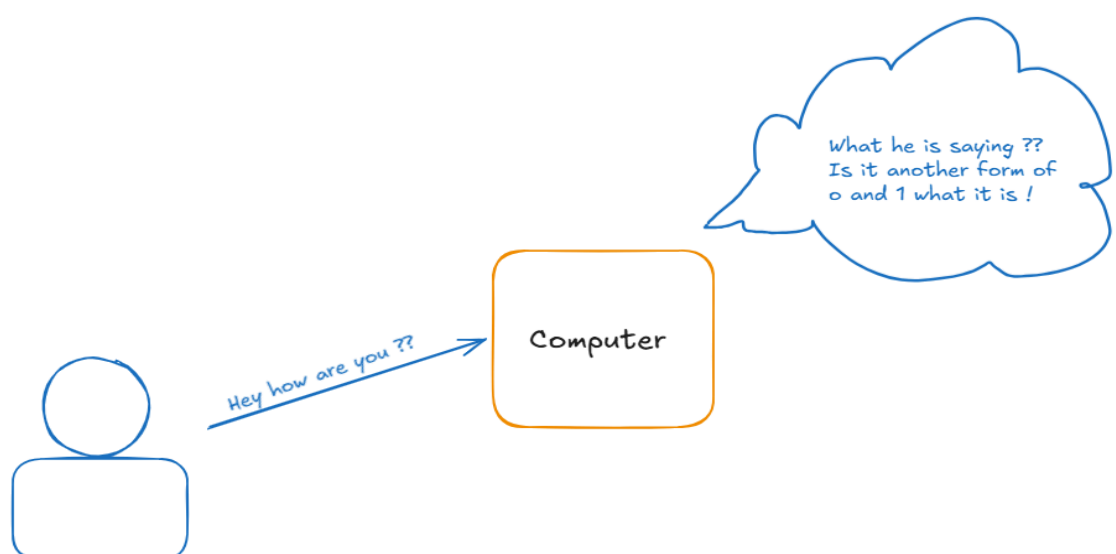
It studies language as data and develops a model that can analyse linguistic structure, meaning and context in both written and spoken communication.

Core Parts of NLP

- **Natural Language Understanding (NLU):** Focuses on helping a computer comprehend the meaning, intent, and context of text or speech.
- **Natural Language Generation (NLG):** Focuses on allowing a computer to produce human-like written or spoken responses

Core problem :

NLP deals with the problem of letting computer understand what is the text , what is the specific language is ! how to talk in Human language .



As we know computers are built on 0's and 1's we cannot work with raw text
So what's the plan ?? how we will let computers understand the language ???

It is simple we can use Linear algebra , probability , stats and python to make it low level to let computer understand and produce the Natural language !!

Where you will find NLP ??

There are tons of NLP applications

- Voice assistants like Alexa, Google Assistant, etc
 - Chatbots in customer support
 - Email spam filtering
 - Auto-correct and predictive typing
 - Language translation tools
 - Social media sentiment tracking
 - Document search and recommendation systems
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Common tasks NLP deals with

- **Text classification:** Assigning predefined labels to text like spam or topic categories.
- **Sentiment analysis:** Detecting whether text expresses positive, negative or neutral emotion.
- **Machine translation:** Automatically converting text from one language to another.
- **Named Entity Recognition:** Identifying names of people, places, dates, etc in text.

- **Text summarization:** Generating a shorter version of a document while keeping key meanings.
- **Question answering systems:** Systems that read text and return exact answers to queries.